

Austin Broderick McElroy

Embedded Software Engineer | Rust Firmware | Medical Devices

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Experience

Senior Staff Software Engineer | Pictorus / Renesas

10/2024 - Present | Remote - Austin, Texas

- Designed, assembled, and brought up an STM32G474 hardware-in-the-loop test PCB covering UART, I2C, SPI, GPIO, ADC, DAC, CAN, and PWM; integrated it into automated nightly end-to-end testing.
- Built and demonstrated a custom no_std Rust HAL for a Renesas MCU in two weeks, covering power, I2C, GPIO, and timers. Renesas later acquired Pictorus. Open sourced as [ra4m2-hal](#).
- Implemented bearer-token-authenticated API endpoints for building, deploying, and modifying firmware programmatically.
- Integrated Rust firmware into customer C and Rust environments, including NuttX/PX4 and FreeRTOS, using static-library FFI interfaces and hands-on hardware debugging.
- Owned a cross-platform CLI for telemetry and OpenOCD firmware updates; improved telemetry bandwidth 4x by replacing JSON with Postcard serialization, writing a Python Postcard decoder for the backend.
- Led delivery of protocol drivers and code generation for PWM, ADC, DAC, and SPI across STM32, Renesas, and embedded Linux targets in Rust, with supporting Python backend and TypeScript frontend changes.
- Collaborated with Ferrous Systems on [pictorus-rs](#), a cross-platform Rust framework for composing reusable components in software and firmware. Open sourced as [pictorus-rs](#).

Staff Scientist | BridgeSource Medical

08/2020 - 10/2024 | Austin, Texas

- Architected firmware and telemetry software for four medical-device SBIR projects totaling \$8M, using embedded Rust, bare-metal C, and Zephyr RTOS across Cortex-M microcontrollers.
- Implemented real-time output and acquisition at 1 MHz for inter-cardiac volume detection using RTIC Rust on an STM32G474, with telemetry to a Tauri analysis and configuration tool.
- Led a \$250k breath-adaptive nebulizer program through successful Phase I human testing, helping secure \$2.5M in Phase II funding; designed its two-stage boost and three-channel piezo drive electronics.
- Built no_std Rust/Embassy firmware for an nRF52840 and a Flutter iOS app over BLE to control haptic shoe orthotics used in a small clinical trial.
- Developed bare-metal C TM4C and Zephyr based nRF52840 firmware with inter-processor communication, BLE device firmware updates, and LVGL-guided device workflows.

CEO / Engineering Consultant | RaptorView, LLC

01/2010 - Present | Austin, Texas

- Designed and brought up a custom ESP32-S3 PCB for a 64x64 RGB [Game of Life matrix](#), including USB-C PD and a 30W buck supply. Firmware written in Rust using ESP-IDF and a reusable no_std simulation core with WASM demo.
- Developed and released [Historicle.ai](#), a Rust / Tauri journaling app with local LLM support using LlamaCPP bindings, backed by SQLite for entries and RAG, with a rich feature set. [Available in the macOS App Store](#).
- Designed and fabricated a USB-C powered LED drive circuit, Rust firmware (nRF52840), and React Native iOS app for a BLE controlled, [LED light engine](#).
- Architected multi-threaded C# data acquisition (DAQ) and display GUI software for a vaginal OCT system and supported the system through clinical trial.
- Designed, simulated, laid out, and manufactured a high-voltage flash-lamp circuit for acne treatment; managed the successful FDA 510(k) application (K111937).
- Developed and open sourced [OpenCLV](#), a LabVIEW package for high-speed computing with OpenCL.

Experience (continued)

Research Scientist | University of Texas - Austin

06/2012 - 05/2020 | Austin, Texas

- Designed the PCB and firmware for **UT ABBU**, an emergency breathing-assist system for hospitals during COVID-19.
- Developed TensorFlow models for OCT segmentation using U-Net and auto-encoders, resulting in two patents.
- Accelerated OCT phase processing for blood-vessel detection 600x with OpenCL.
- Designed and built custom analog photodetectors up to 400 MHz for laser imaging systems.
- Led LabVIEW DAQ and OpenCL image processing for five OCT platforms, supporting dozens of publications.
- Architected laser sintering 3D printer software integrating FPGAs, microcontrollers, and high-speed thermal imaging; helped secure \$2M in Phase II funding.

Technical Skills

Languages: Rust, C, Python, TypeScript, LabVIEW.

Embedded: no_std, Embassy, ESP-IDF, RTIC, Zephyr, FreeRTOS, LVGL, Cortex-M, embedded Linux.

Hardware: Board bring-up, HIL, JTAG, OpenOCD, oscilloscopes, logic analyzers, hardware debugging.

Tools: Git, Cargo, CMake, KiCad, SPICE, Yocto, Docker, unit/integration tests, DAQmx.

Education

Master of Science in Electrical Engineering | University of Texas, Austin

Bachelor of Science in Electrical Engineering | University of Texas, Austin

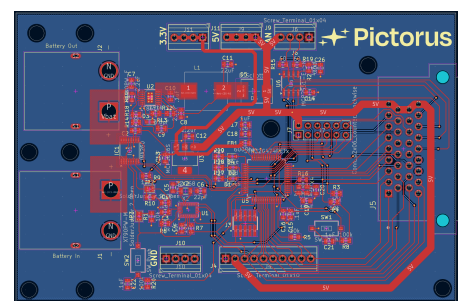
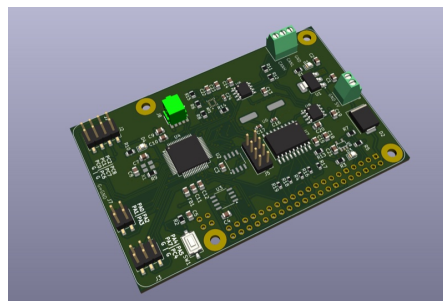
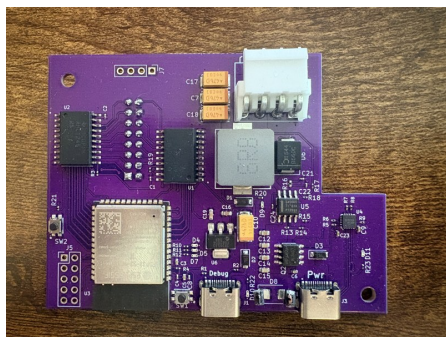
Selected Patents (3 of 6)

- Multi-channel orthogonal convolutional neural networks. US Patent 11,779,220 (2023).
- Real time SD-OCT with distributed acquisition and processing. US Patent 9,347,765 (2016).
- Apparatus and methods for uniform frequency sample clocking. US Patent 8,049,900 (2011).

Selected Publications (3 of 72)

- Katta et al. "Laser brain cancer surgery in a xenograft model guided by optical coherence tomography." *Theranostics* 9(12), 3555 (2019).
- Katta et al. "Optical coherence tomography image-guided smart laser knife for surgery." *Lasers in Surgery and Medicine* 50(3), 202-212 (2018).
- Wang et al. "Dual-modality fiber-based OCT-TPL imaging system for simultaneous microstructural and molecular analysis of atherosclerotic plaques." *Biomedical Optics Express* 6(5), 1665-1678 (2015).

Selected Embedded Systems



Left to right: ESP32-S3 Game of Life board; STM32G474 hardware-in-the-loop board; CBot PCB layout.